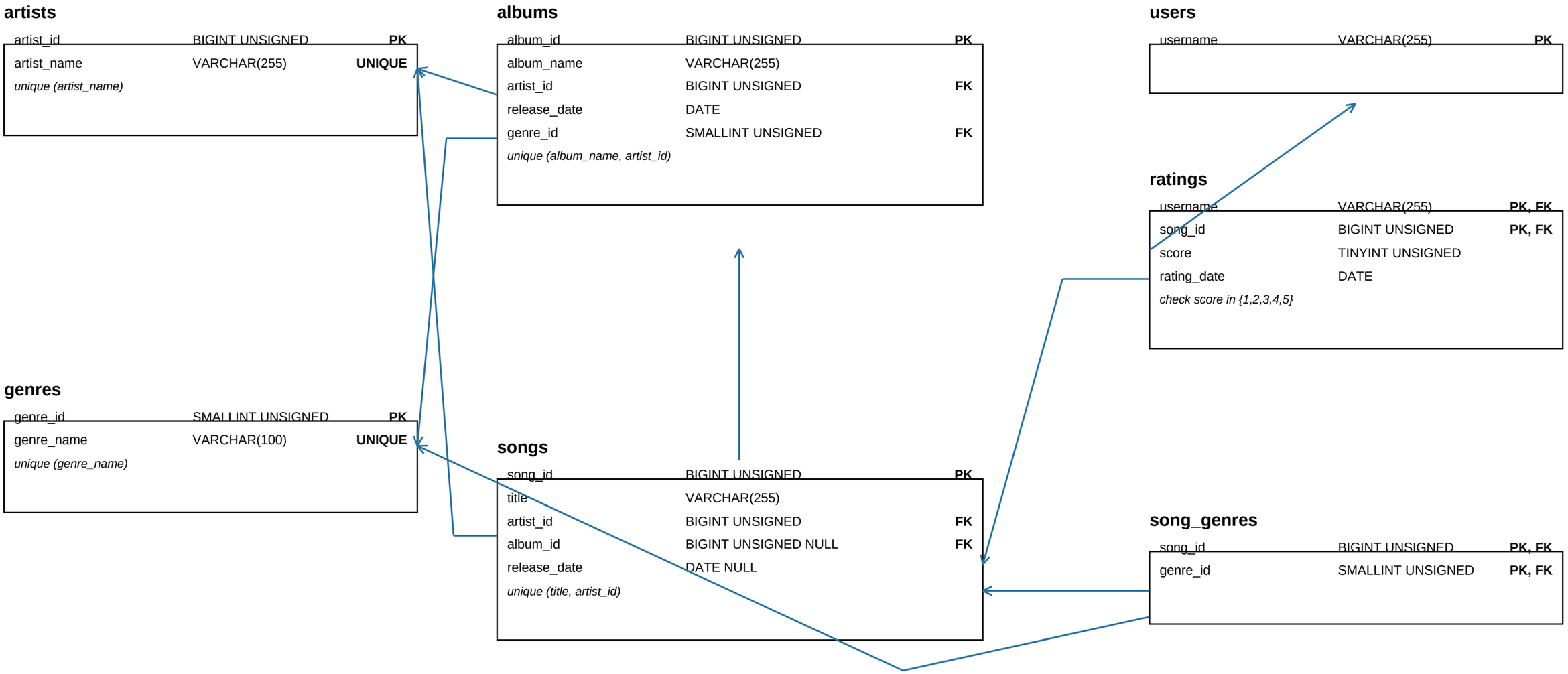
Music Database Schema

Relational schema for artists, albums, songs, genres, users, and ratings



Design notes

- Artist names and usernames are unique as required; surrogate integer keys reduce repeated storage in child tables.
- Album names are not globally unique, but album_name plus artist_id is unique.
- A song title may be shared by different artists, but title plus artist_id is unique.
- Single songs store their own release_date; album songs point to albums and inherit the album release_date.
- song_genres supports one or more genres per song; the Python loader enforces at least one genre.
- Album songs receive the album genre through song_genres, keeping genre queries uniform for singles and album tracks.

Foreign key arrows

albums.artist_id -> artists.artist_id
albums.genre_id -> genres.genre_id
songs.artist_id -> artists.artist_id
songs.album_id -> albums.album_id
ratings.username -> users.username
ratings.song_id -> songs.song_id
song_genres.song_id -> songs.song_id
song_genres.genre_id -> genres.genre_id